229 Basic Idea of Automatic Knowledge Retrieval System

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1 Introduction

In spite of the years of research in the field of Artificial Intelligence, the creation of a machine with the ability to think is still far from realization. Although computer systems are capable to perform several complicated tasks that require human beings to extensively use their intelligence and thinking capabilities, still machines cannot understand what people talk about, why do they get angry or happy. One of the main, unresolved problem is the lack of commonsensical knowledge.

Some researches estimate that human commonsensical knowledge consists of approximately 1,000,000,000 bits (Landauer 1986).In the last decades we could have witnessed a few attempts to create such a database manually. As we can learn from the experiences of these projects, building a database by inputting the knowledge concepts is laborious, time-consuming and costly process. We argue that in the open domain where new information appears and becomes obsolete on a daily basis, a complete knowledge base build manually is simply out of reach. Comparing to the efforts taken to input the knowledge concepts manually, surprisingly little has been done to develop the methods for the automatic creation of such databases. Below we present the idea of the language independent method capable to automatically retrieve general and commonsensical knowledge for the open-domain applications.

2 Knowledge for the Machine

The attempts to manually construct a knowledge database has been made in a few ongoing projects like: Cyc (Lenat) or Thought Treasure (Mueller). These projects concentrate on creating databases, which could provide knowledge from possibly large scope of domains, along with the means to effectively use it for the systems that engage in the reasoning about human affairs. However, the manual approach to construct such a database require enormous amount of time, human labor and money.

In recent years, researches have demonstrated that it is possible to populate knowledge databases directly, using normal means of communication such as written natural language (Katz 1995), and that the items imputed in this way can be used for reasoning (Borchardt 1992, Singh 2002). This approach was used in construction of the Open Mind Common Sense database (OMCS), which simplified the process of database building to the point where literally every Internet user without any special training could input knowledge concepts using plain sentences written in English. While decreasing the amount of money spend, this approach still requires large investment of human labor and time for creation of the knowledge database.

The aim of the method described below is to automatically retrieve the knowledge concepts that form valid entries to a database. This method differs from the knowledge acquisition, which is defined as the process of extracting, structuring and organizing knowledge from several sources, usually human domain experts, so it can be used in a program (Waterman 1985). Some relevant research, however adapting different approaches and to some extent having different aims include the following works. Studies on knowledge representation and classification were done by Woods, followed by the work on automatic construction of the taxonomies by extracting concepts from texts [4]. The research of Rzepka [3] describes the methods for automatic word categorization and script generation using the statistic of words obtained from the Internet and by exploring the Japanese grammar features. In the most closely related work [2], Moldovan described the knowledge acquisition system that gathers information from the financial domain. However, this system is not only tailored to a specific field, its processing can not be done automatically, as the system requires an interaction with a user to verify the acquired concepts, relationships and extraction patterns.

3 Automatic Knowledge Retrieval System

In our opinion the Internet is a rich resource of commonsensical and general knowledge that can be used for the automatic creation of the knowledge databases. For the automatic knowledge retrieval this resource has important advantages including the real-time updating of its content, wide coverage of various domains and the diversity of presented opinions. The difficulties that need to be addressed when using this resource include the need to filter out statements that cannot be considered as the reliable and valid entries to a knowledge database as well as overrepresented commercial information. The challenge here is to ensure a high precision of concept selection while obtaining a wide coverage of various domains and high recall rate of knowledge concepts.

Below we present "KnowY" system, designed to automatically retrieve the knowledge concepts from the Internet to create a database similar to the ones that are being built manually, described in **2**. At present, the system utilizes the OMCS database to learn what constitutes a valid entry to a knowledge database and to automatically rank the concepts depending on their similarity to the ones found in the manually constructed knowledge database.

4 System Processing

"KnowY" is capable to work both in offline and online mode. In the former one the system automatically creates new entries to a knowledge base, while the online mode is designed to serve as a helper application for the dialog system (the initial implementation is for the ALICE chattering-bot) by providing it with the knowledge related to a discussed topic. In this mode "KnowY" uses additional features that provide faster response time by retrieving only this knowledge which is required by the application in a given situation and related to a dialog domain. To present these additional features we decided to describe the system processing when used in the online mode. Figure 1 presents the system flowchart.

4.1 Domain Recognition

In the online version "KnowY" interacts with the dialog system, and is initiated by receiving a request for additional knowledge on a given noun. As we know, nouns frequently have several distinct senses, are related to different domains, and it is often difficult to decide beforehand which one is the most suitable in a given situation. If this information can-

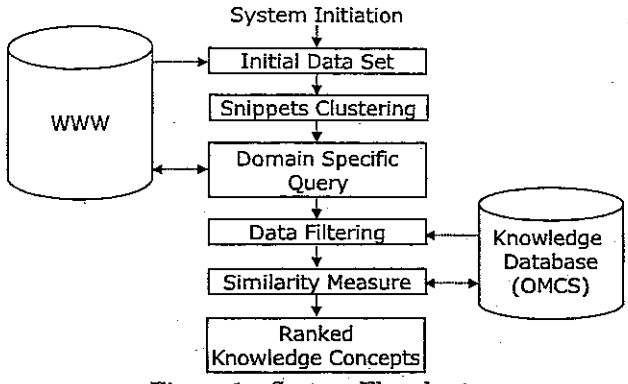


Figure 1　System Flowchart

not by acquired from the user utterance to the dialog system, "KnowY" tries to obtain a diverse set of information related to a requested noun where a word sense/domain intended by a user can be found. To provide such a set, the system retrieves 1500 highest scored snippets from 5 different domains (.gov .edu .net .org .com). Next, a set of the unique snippets is clustered [1], and for each cluster 3 most descriptive words are discovered. Using a query composed of these words and the requested noun "KnowY" searches the Internet and counts the frequency of the statements that consists the following patterns: "noun[s] $\langle BE \rangle$ $\langle a/the/\ /\rangle$ $\langle descriptive\ word_{noun[s]} \rangle$" or "noun[s] $\langle BE \rangle$ $\langle descriptive\ word_{adjective/adverb} \rangle$". The most frequent statements are used to form questions in the natural language, which are returned to the dialog system to find a domain intended by a user. Once this information is obtained "KnowY" executes the new search using a query consisting of a requested noun along with the 3 most descriptive words discovered for a selected cluster (domain). This time the system retrieves a set of 300 web pages, which ensures a wide selection of information concerning a noun while respecting the domain limitation. After dividing the text into sentences, the ones that do not include the requested noun (either in singular or plural form) are excluded. For example, for the noun "apple" within the domain (tree, fruit, apples) a set of 4,140 sentences was obtained, with 2,751 unique ones (the concurrence of the sentences is used later in the system processing to assign the commonness rate to the knowledge concepts).

4.2 Automatic Knowledge Concepts Discovery

In the data filtering and similarity measure stages, the OMCS database is used as training data that reveal the most frequent ways of describing knowledge concepts as well as commonly used words and grammatical constructions. To ensure reasonable processing time the system uses only a set of concepts from the OMCS database. This set is composed of the concepts that include the requested noun (either in singular or plural form). If the number of such concepts is less than 100, the system randomly adds additional ones to obtain a set of 1,000 concepts. Obviously for a well defined concepts (high quality and wide coverage of a given noun in the OMCS in the database), the former method is likely to find a better set of automatically retrieved concepts. On the other hand, the later approach provides the means to discover the new concepts, and allows "KnowY" to generate a database for a terms which are not covered by the OMCS at all. We think that this feature is of prime importance for many applications in the open domain. The average sentence length found in the OMCS database was 9.32 words long. Considering the time processing requirements, for the similarity score ranking we decided to include only the "web sentences" and OMCS concepts with the number of words between 3 and 20.
To exclude the sentences that are unlikely to be valid entries to the knowledge database, "KnowY" compares them to the OMCS knowledge database concepts and ranks using the highest similarity score obtained, calculated with the following formula:

$$Similarity = \frac{\sum_{n=1}^{N} Wn * (Wn_{ITF})}{L1 + L2} + \log(Np) * \alpha \quad (1)$$

,where: Wn - matching word found both in a OMCS concept and a "web sentence", Wn_{ITF} - inverted term frequency for a matching word (OMCS), L1 - number of words in a "web sentence", L2 - number of words in a OMCS concept, Np - number of concepts from OMCS where a noun was found in a position from a "web sentence", α - parameter (the value of α was set to 0.2). The used formula takes account of the number of matching words between a "web sentence" and a OMCS concept as well as the importance of the given word in a used set of OMCS by the means of the ITF. The Np value, promotes "web sentences" where a noun appears on a position, which is frequent for many OMCS concepts.

5 Conclusions and Future Work

We think that the attempts to create knowledge databases manually provide many important experiences that we can learn from as well as valuable resources useful for several applications in the field of AI. However as explained above, the manual construction of a complete knowledge database for an open-domain is not feasible. This paper presented the method of automatic knowledge retrieval system. We think that the idea described here and implemented in "KnowY" contributes to address one of the most important challenge in the AI that exists nowadays.
Our future work includes further refining of the presented method and the application of other knowledge databases for creating a comparison set for the data filtering and similarity measurements. Since the time requirements of the offline version of "KnowY" are not as strict as the online one, we consider also extending it with the additional processing, including the syntax analysis for a similarity measurements, which should be especially beneficial in the cases where a OMCS comparison set does not include sufficient number of concepts for a requested noun. Given that the described method is language independent we intend also to evaluate it using languages other than English.

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